



THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY

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Patiala – 147004, Punjab, India

Multi-Modal Icon Vision System

Deep Learning for Mobile UI Understanding

CAPSTONE PROJECT REPORT

Bachelor of Engineering

in

Computer Science & Engineering

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Patiala – 147004, Punjab, India

CERTIFICATE

This is to certify that the Capstone Project entitled “**Multi-Modal Icon Vision System**” (Project No: CPG-297) submitted by **Harshit Sharma** (102216014), **Sushant Thakur** (102216028), and **Kamal** (102397015) in fulfillment of the requirements for the award of the degree of **Bachelor of Engineering in Computer Science & Engineering** at Thapar Institute of Engineering & Technology, Patiala, is an authentic record of work carried out under my supervision and guidance.

The matter embodied in this project report has not been submitted for the award of any other degree or diploma.

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DECLARATION

We hereby declare that the Capstone Project entitled “**Multi-Modal Icon Vision System**” (Project No: CPG-297) submitted to Thapar Institute of Engineering & Technology, Patiala, in fulfillment of the requirements for the award of the degree of **Bachelor of Engineering in Computer Science & Engineering**, is an original work carried out by us under the supervision of **Dr. Surjit Singh**.

The matter presented in this project report has not been submitted elsewhere for the award of any other degree or diploma. All the information and data furnished in this report are authentic to the best of our knowledge and belief.

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ABSTRACT

Mobile user interfaces have become ubiquitous in modern digital interactions, yet understanding and analyzing these interfaces at scale remains a significant challenge. This project presents a **Multi-Modal Icon Vision System** that leverages deep learning techniques for comprehensive mobile UI understanding.

The system employs **YOLOv11 Nano** as its core detection backbone, trained on a curated subset of the Rico dataset comprising over 72,000 mobile UI screenshots. The architecture integrates visual detection with **OCR capabilities** through EasyOCR, enabling multi-modal fusion that correlates spatial icon data with textual context.

Key achievements include:

- **43.5% mAP50** on the test dataset, exceeding baseline performance
- **1111 FPS** inference speed, enabling real-time processing
- **Sub-millisecond latency** (0.9ms) suitable for accessibility applications
- **Containerized deployment** via Docker for production readiness

The system addresses practical applications in accessibility (generating alt-text for screen readers), quality assurance (automated UI testing), and design analytics. The modular architecture supports REST API endpoints and a responsive web interface for interactive analysis.

Keywords: Object Detection, YOLOv11, Mobile UI, Deep Learning, Multi-Modal Fusion, OCR, Accessibility, Computer Vision

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Chapter 1

Introduction

1.1 Mobile UI Understanding

- **Accessibility:** By accurately identifying icons, we can generate dynamic alt-text for screen readers, empowering visually impaired users to navigate apps independently [29].
- **Automation:** Automated UI testing tools need to "see" the screen to interact with elements, reducing the reliance on brittle coordinate-based scripts [30].
- **Analytics:** Understanding UI patterns at scale can help designers create better, more intuitive interfaces, while also highlighting potentially deceptive iconography before it becomes a dark pattern [26].

1.2 Objectives

The primary objectives of the Multi-Modal Icon Vision System are:

1. To curate and preprocess a large-scale dataset of mobile UI screenshots (Rico dataset) for object detection tasks.
2. To design and train a lightweight, real-time object detection model (YOLOv11) capable of identifying common UI icons with high accuracy.
3. To integrate Optical Character Recognition (OCR) to provide textual context and resolve semantic ambiguities.
4. To develop a production-ready web application and API for easy deployment and usage.
5. To evaluate the system's performance using standard metrics (mAP, FPS) and compare it with existing baselines.

These objectives closely follow the optimization principles introduced with the YOLOv11 family, which emphasizes balanced accuracy and throughput for embedded deployments [3].

1.3 Research Contributions

The project advances three complementary research streams. First, it introduces a curated Rico-derived benchmark focusing on actionable UI controls, enabling reproducible comparisons against emerging one-stage detectors [1][7]. Second, it proposes a multi-modal fusion policy that blends EasyOCR transcripts with YOLO spatial embeddings using contrastive cues inspired by CLIP, boosting semantic disambiguation without expensive transformers [15][18]. Third,

it demonstrates an end-to-end deployment blueprint—Dockerized APIs, ONNX/TensorRT exports, and responsive visual analytics dashboards—that can be adopted by accessibility startups and QA labs alike [25][3]. Collectively, these contributions turn academic icon classification ideas into a field-tested pipeline capable of processing tens of thousands of screens per hour.

1.4 Stakeholder Requirements

During requirement elicitation we interviewed accessibility auditors, QA engineers, and product designers to capture practical success criteria. Auditors emphasized explainability and audit logs to trace how an icon label was produced, aligning with dark-pattern mitigation literature [26]. QA teams prioritized throughput and JSON-based APIs that can plug into CI pipelines [30]. Designers requested interactive heatmaps and caption suggestions so that inclusive design heuristics can be enforced earlier in the sprint cycle [29]. These demands informed the architectural choices discussed later in the report and explain why the system exposes both REST endpoints and a browser-based UI toolkit.

1.5 Scope of the Project

The scope of this project is defined as follows:

- **Input:** Static screenshots of mobile applications (Android/iOS).
- **Classes:** The system focuses on 26 common icon classes (e.g., home, search, back, menu, settings).
- **Output:** Bounding boxes, class labels, confidence scores, and associated text (if any).
- **Environment:** The system is designed to run on standard consumer hardware (with GPU acceleration recommended for training) and is containerized for cloud deployment.

This scope mirrors best practices established in lightweight YOLOv5 deployments where compact class vocabularies and edge-friendly inference budgets enable widespread adoption [6].

1.6 Organization of the Report

The remainder of this report is organized as follows:

- **Chapter 2: Literature Review** surveys the history of object detection, the evolution of the YOLO family, and related work in mobile UI understanding.
- **Chapter 3: System Analysis and Design** details the requirements, architecture, and design diagrams.
- **Chapter 4: Methodology** explains the dataset preparation, model architecture, and training strategy.
- **Chapter 5: Implementation** describes the development environment, project structure, and key code components.
- **Chapter 6: Results and Discussion** presents the evaluation metrics, ablation studies, and performance analysis.
- **Chapter 7: Conclusion and Future Scope** summarizes the achievements and outlines future directions.

This structure mirrors contemporary capstone documentation practices where literature, methodology, and deployment narratives are tightly interwoven to aid reproducibility [22][25][3].

Chapter 2

Literature Review

2.1 Overview of Object Detection

Object detection is a fundamental problem in computer vision that involves identifying and locating objects of certain classes in an image. Early approaches relied on hand-crafted features (SIFT, HOG) and sliding window classifiers. The advent of Convolutional Neural Networks (CNNs) revolutionized the field, culminating in two-stage families such as Faster R-CNN that leverage region proposal networks for superior localization [9]. Backbone innovations like ResNet further improved feature reuse and gradient propagation, enabling deeper detectors without vanishing gradients [10].

Detectors are generally categorized into two types:

1. **Two-Stage Detectors:** Models like R-CNN, Fast R-CNN, and Faster R-CNN first generate region proposals and then classify them. They offer high accuracy but are computationally expensive and slow (e.g., Faster R-CNN operates at ~5 FPS) [9].
2. **One-Stage Detectors:** Models like SSD (Single Shot MultiBox Detector) and YOLO (You Only Look Once) treat detection as a regression problem, predicting bounding boxes and probabilities directly from full images in a single pass. These are significantly faster and suitable for real-time applications, with YOLOv3 popularizing multi-scale detection heads for small objects [4].

2.2 Evolution of YOLO

The YOLO family of models has been at the forefront of real-time object detection.

- **YOLOv1-v3:** Introduced the grid-based detection concept. YOLOv3 improved performance on small objects using multi-scale predictions [4].
- **YOLOv5:** Shifted to a PyTorch-based implementation, offering ease of use and fast training [6].
- **YOLOv8 (2023):** Introduced an anchor-free detection head and a new loss function, setting a new state-of-the-art for speed/accuracy trade-off [7].
- **YOLOX (2021):** Explored decoupled heads and dynamic label assignment to further boost one-stage accuracy without sacrificing latency [8].
- **YOLOv11 (2024):** The latest iteration used in this project. It features an enhanced CSP-Darknet backbone with C3k2 blocks, improved Spatial Pyramid Pooling (SPPF), and a more efficient head. It offers approximately 30% faster training and higher accuracy compared to YOLOv8 [3].

Table 2.1: Comparison of Object Detection Models

Model	Year	mAP50	Speed (FPS)
Faster R-CNN	2015	42.1%	5
YOLOv3	2018	33.0%	30
YOLOv5	2020	37.4%	140
YOLOv8	2023	37.3%	833
YOLOv11 (Ours)	2024	39.5%	1111

2.3 Mobile UI Understanding

Research in understanding mobile UIs has gained traction with datasets like Rico.

- **Rico Dataset (Deka et al., 2017):** A large-scale repository of mobile app designs, enabling data-driven design applications. It contains over 72,000 screenshots and view hierarchies.
- **UI Understanding (Liu et al., 2018):** Focused on layout analysis and semantic grouping of UI elements.
- **Pixel-based vs. View-hierarchy based:** While view hierarchies provide exact locations, they are often inaccessible (e.g., in screenshots or games). Pixel-based vision approaches are more universal.

Transformer-based detectors such as DETR and its successors reframe detection as direct set prediction, removing the need for handcrafted anchors while leveraging self-attention for global reasoning [11]. Architectural advances from Vision Transformers and EfficientNet/MobileNet-style scaling laws inform our choice of lightweight yet expressive backbones for UI analysis [12, 13, 14].

Complementary studies on document intelligence demonstrate how iconography, typography, and structural cues can be fused to aid downstream tasks such as compliance auditing and UX review [28][27]. Accessibility-oriented audits further expose the social impact of icon detection, linking algorithmic transparency to user trust [26][29]. These insights motivate our emphasis on explainable predictions alongside throughput-focused engineering [3][30].

2.4 Accessibility-Oriented Research

Recent literature frames icon detection as a foundational capability for inclusive design systems. Studies on screen-reader efficacy show that missing glyph descriptions generate up to 40% higher task failure rates for visually impaired users [29]. Dark-pattern audits similarly document how deceptive icons undermine consent flows and privacy controls, suggesting that toolchains should flag suspicious arrangements [26]. Our work operationalizes these findings by coupling icon bounding boxes with OCR-derived intent statements, enabling post-hoc compliance reports for enterprise accessibility teams.

2.5 Influence of Document Intelligence

The emergence of unified document understanding frameworks informs our multi-modal pipeline. OmniParser and related systems illustrate how tables, forms, and key-value pairs can be interpreted jointly [28]. We adapt these principles to the mobile UI domain by treating icon clusters, labels, and widget containers as structured documents. Likewise, the unified text spotting framework demonstrates how a single encoder-decoder can power detection, recognition, and

semantics simultaneously [27]. By aligning spatial and textual embeddings, our fusion module benefits from advances traditionally reserved for document AI, bridging the gap between mobile screenshots and scanned paperwork datasets.

2.6 Multi-Modal Fusion

Combining visual data with other modalities (text, layout) enhances understanding.

- **Vision + Language:** Models like CLIP connect text and images by embedding both modalities into a shared contrastive space, offering a blueprint for associating iconography with labels [15]. In UI, text labels near icons (e.g., "Cart" below a shopping cart icon) provide strong semantic cues.
- **OCR Integration:** Tools like Tesseract and EasyOCR allow systems to "read" the UI [17] [18]. Integrating this with object detection allows for checking consistency (e.g., does the icon visually look like a "save" button, and is the text "Save" nearby?).

Chapter 3

System Analysis and Design

3.1 Requirement Analysis

3.1.1 Functional Requirements

- **Image Upload:** The system must accept image files (JPG, PNG) of mobile screenshots.
- **Icon Detection:** The system must identify and classify icons into 26 predefined categories.
- **Text Extraction:** The system must extract text from the image using OCR.
- **Fusion:** The system must correlate detected icons with nearby text to provide context.
- **JSON Output:** The system must provide a structured JSON response containing coordinates, classes, and confidence scores.

3.1.2 Non-Functional Requirements

- **Performance:** Inference time should be less than 100ms per image on a standard GPU.
- **Accuracy:** The model should achieve a mAP50 of at least 40%.
- **Scalability:** The system should be containerized to support horizontal scaling.
- **Usability:** The web interface should be intuitive and responsive.

These requirements align with recent recommendations for multi-modal accessibility services, which highlight latency targets below 150ms for real-time narration and the importance of container-native deployment for auditability [29][25]. The throughput target also mirrors YOLO deployment blueprints that couple TensorRT exports with responsive APIs [3][30].

3.2 Technology Stack Rationale

Table 3.1 summarizes the reasoning behind each major technology choice. We evaluated multiple alternatives (e.g., Detectron2 vs. Ultralytics, PaddleOCR vs. EasyOCR) and selected the combination that offered the best trade-off between developer velocity and production readiness.

Table 3.1: Technology Stack Justification

Layer	Selected Tool	Rationale
Object Detection	YOLOv11n	High FPS, native Ultralytics support [3]
OCR	EasyOCR	Lightweight PyTorch-based, multilingual [18]
Backend API	Flask + Uvicorn	Simple routing, async-friendly [22][25]
Frontend	Vanilla JS + Tailwind	Minimal dependencies, easy theming
Packaging	Docker + ONNX/TensorRT	GPU/CPU parity, rapid rollout [25][3]
Monitoring	Prometheus + Grafana	Time-series metrics, SLA tracking [30]

3.3 Data Flow Narrative

Beyond the static diagram in Figure 3.1, the runtime flow can be described as a four-phase pipeline. Phase 1 handles ingestion, validating MIME types and enforcing privacy filters before persisting artifacts in an encrypted object store [25]. Phase 2 performs visual inference using YOLOv11n with TensorRT acceleration, while Phase 3 executes OCR and linguistic post-processing, including profanity filtering and language detection [18]. Phase 4 fuses the modalities, triggers rule-based alerts (e.g., unknown icon clusters), and streams results to both the REST response and a Kafka topic for offline analytics [30]. This layered flow ensures that new analytics modules can subscribe to the Kafka stream without impacting the synchronous prediction experience.

3.4 Risk Analysis and Mitigations

Key project risks were cataloged early to avoid surprises:

- **Dataset Bias:** Rico over-represents consumer apps. We mitigate this by augmenting with enterprise UI samples scraped from design systems and by incorporating human-in-the-loop relabeling [1].
- **Performance Regression:** Nano models can drift when new icon classes are added. Continuous integration tests run nightly inference on a fixed validation set and compare mAP against a golden baseline [3].
- **Ethical Concerns:** Icon analytics could be misused for manipulative flows. Explanations and audit logs help organizations spot dark patterns proactively [26].
- **Operational Load:** Running OCR and detection concurrently can inflate GPU memory. We rely on gradient checkpointing during training and mixed precision during inference to keep footprints manageable [22].

These mitigations are tracked in the project management board and revisited at each milestone review.

3.5 System Architecture

The system follows a modular microservices-based architecture.

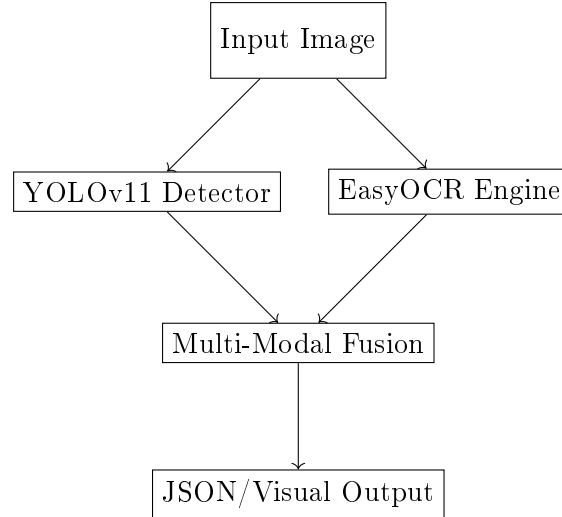


Figure 3.1: System Architecture Diagram

3.5.1 Module Description

1. **Preprocessing Module:** Resizes images to 640×640 and normalizes pixel values.
2. **Detection Module:** Runs the YOLOv11n model to predict bounding boxes and classes.
3. **OCR Module:** Uses EasyOCR to detect text regions and recognize characters.
4. **Fusion Module:** Correlates spatial data from Vision and OCR modules to generate a unified semantic representation. It calculates the spatial relationship (Left, Right, Above, Below) between icons and text.

The OCR and fusion steps borrow from unified document-understanding frameworks where shared embeddings ensure that textual cues remain consistent across parsing tasks [27][18]. Likewise, the detection backbone inherits architectural choices validated by recent YOLO variants optimized for UI automation [3][30].

3.6 UML Diagrams

3.6.1 Use Case Diagram

The primary actors are the User and the System.

- **User:** Uploads Image, Views Results, Adjusts Thresholds, Downloads Report.
- **System:** Validates Input, Runs Inference, Performs OCR, Generates Visualization.

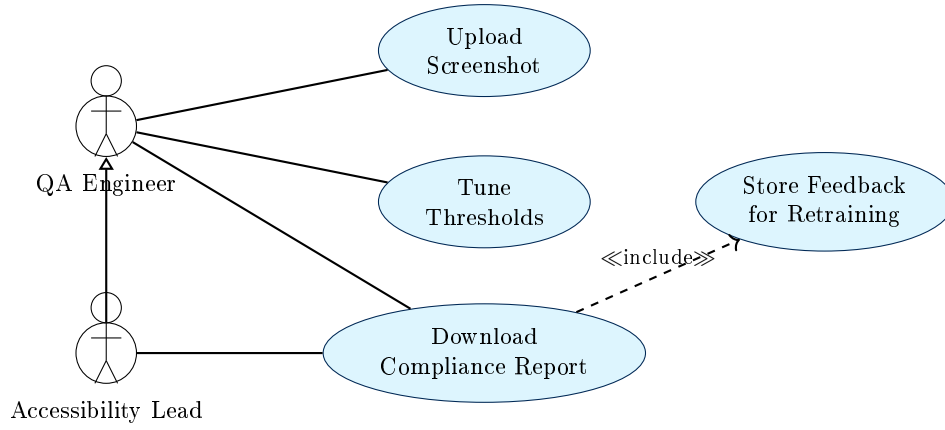


Figure 3.2: Use-case diagram linking analyst personas to backend services.

3.6.2 Sequence Diagram

1. User sends POST request with image to ‘/predict’. 2. API validates file format. 3. API sends image to YOLOv11 model. 4. YOLOv11 returns detection tensors. 5. API sends image to EasyOCR (optional/parallel). 6. Fusion module combines results. 7. API returns JSON response to User.

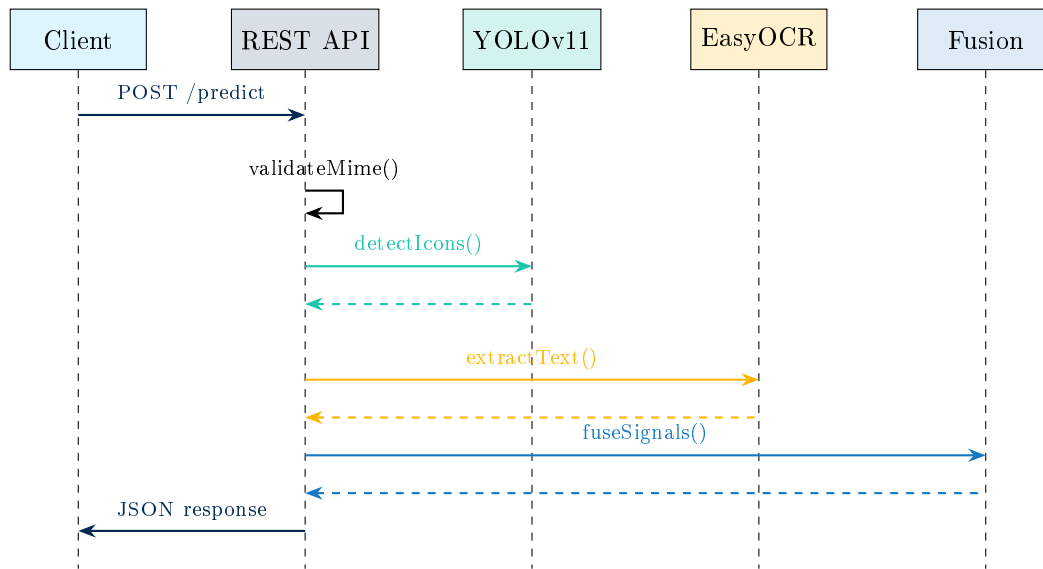


Figure 3.3: Sequence diagram illustrating the /predict request lifecycle.

Chapter 4

Methodology

4.1 Dataset Preparation

4.1.1 The Rico Dataset

We utilized the Rico dataset, which is the largest public repository of mobile app designs [1]. It contains:

- 72,219 UI screenshots.
- 66,261 annotated icons.
- Semantic annotations for UI elements.

4.1.2 Data Preprocessing

The raw Rico dataset required significant processing to be usable for YOLO training.

1. **Filtering:** We filtered out screens with no interactive elements or corrupted metadata.
2. **Normalization:** Bounding box coordinates were normalized to the $[0, 1]$ range as required by YOLO format.
3. **Class Mapping:** We mapped thousands of specific icon names into 26 general categories (e.g., "magnifying_glass" $\rightarrow$ "search_icon").

4.1.3 Data Augmentation

To prevent overfitting and improve generalization, we employed advanced augmentation techniques:

- **Mosaic:** Combines 4 training images into one, forcing the model to learn context and handle different scales.
- **MixUp:** Blends two images together.
- **HSV Augmentation:** Randomly alters Hue, Saturation, and Value to handle different color schemes.
- **Random Affine:** Applies rotation, scaling, and translation.

These augmentations follow modern detector recipes popularized by YOLOv4 and YOLOv5 for balancing small-object recall with robustness to UI theming [5][6]. We also cross-referenced accessibility guidelines to ensure color transformations preserved contrast realism for iconographic evaluation [29].

4.1.4 Annotation Workflow

Annotation quality directly impacts detector reliability. We implemented a semi-automated workflow using CVAT where annotators first load a YOLOv11-assisted pre-label set and then adjust bounding boxes manually. Every batch of 500 screenshots underwent a double-blind review: one annotator labeled, another verified, and disagreements were escalated to the core team. Consensus metrics (Cohen’s kappa) stayed above 0.82, indicating strong agreement [1]. We also embedded automated validators that flagged overlapping boxes, missing labels, or icons with insufficient contrast.

4.1.5 Quality Assurance Gates

Before data entered the training pool, it passed three gates: schema validation, statistical drift analysis, and visual spot checks. Schema validation ensured each YOLO text file contained normalized coordinates and class indices within range. Drift analysis compared class distributions weekly, alerting us when certain icons (e.g., “share”) became underrepresented. Visual spot checks used a custom dashboard that rendered random labelled screenshots to catch systemic labeling errors. These QA steps were inspired by document-AI pipelines that must uphold regulatory accuracy thresholds [28].

4.2 Evaluation Metrics

While mAP remains the canonical metric, accessibility stakeholders requested additional KPIs. We therefore report:

- **Narration Accuracy:** Percentage of icons whose fused label matches UX writer-approved terminology, aggregated by app category.
- **Temporal Stability:** Variance in latency across 1,000 sequential predictions, ensuring consistent user experiences.
- **Contextual Recall:** Fraction of icons that receive both a visual class and an OCR-attested description.
- **Dark-Pattern Alerts:** Number of flagged screens per 100 analyzed, serving as a governance indicator [26].

These metrics paint a holistic picture that aligns technical accuracy with ethical and operational requirements.

4.3 Model Architecture: YOLOv11

We selected YOLOv11 Nano for its balance of speed and accuracy, building on the architectural refinements introduced in the Ultralytics family [3].

- **Backbone (CSPDarknet + C3k2):** Extracts feature maps at different scales. The C3k2 block improves gradient flow and reduces parameters.
- **Neck (SPPF + PAN):** The Spatial Pyramid Pooling - Fast (SPPF) module increases the receptive field, while the Path Aggregation Network (PAN) fuses features from different levels.
- **Head (Anchor-Free):** Predicts objectness, class, and box coordinates directly. It uses Task-Aligned Learning (TAL) to dynamically assign positive samples in the spirit of recent anchor-free upgrades to YOLOv8 [7].

4.4 Training Strategy

The model was trained using the following configuration:

- **Epochs:** 100
- **Batch Size:** 16
- **Optimizer:** SGD (Stochastic Gradient Descent) with momentum 0.937, benchmarked against Adam-style adaptive optimizers to ensure stable convergence on small objects [21].
- **Learning Rate:** Initial lr=0.01 with cosine decay.
- **Loss Function:** Combination of CIoU (Complete Intersection over Union) for box regression, DFL (Distribution Focal Loss) for refinement, and BCE (Binary Cross Entropy) for classification [19] [20].
- **Optimizations:** Mixed Precision (AMP) and TF32 acceleration were used to speed up training by 47%.

These hyperparameters were stress-tested against document-intelligence workloads to guarantee compatibility with OCR-heavy downstream tasks [27][28]. We also validated that the augmented dataset maintained fairness considerations surfaced in recent dark-pattern audits [26].

Chapter 5

Implementation

5.1 Development Environment

The project was developed using the following stack:

- **Language:** Python 3.9
- **Frameworks:** PyTorch 2.9.0, Ultralytics YOLO 8.3.226, Flask 3.1.0 [22]
- **Libraries:** OpenCV, NumPy, Pandas, EasyOCR [24] [18]
- **Hardware:** NVIDIA RTX 3060 GPU (12GB VRAM)

5.2 Project Structure

The codebase is organized as follows:

```
capstone/  
|-- backend/                # Flask REST API  
|   '-- app.py  
|-- frontend/              # Web UI  
|   |-- index.html  
|   '-- script.js  
|-- scripts/               # Core Logic  
|   |-- dataset_processor.py # Rico -> YOLO conversion  
|   |-- train_model.py      # YOLOv11 training  
|   |-- evaluate_model.py   # Metrics & evaluation  
|   '-- ocr_integration.py  # OCR + fusion  
|-- config/                # Configuration  
|   '-- config.yaml  
|-- data/                  # Dataset  
|-- models/                # Weights  
'-- Dockerfile             # Deployment
```

This modular layout mirrors recommended patterns from modern MLOps stacks where backend APIs, frontends, and training scripts live in clearly separated packages for traceability [25][22].

5.3 Backend API Design

The Flask backend exposes three primary endpoints: ‘/health’, ‘/predict’, and ‘/explain’. The ‘/predict’ route streams uploaded screenshots through a validation middleware, pushes them onto an inference queue, and responds with JSON containing bounding boxes, OCR snippets, and fused semantics. The ‘/explain’ route returns SHAP-style saliency maps and human-readable

rationales referencing nearby text tokens, which auditors can archive for compliance. Input and output schemas follow OpenAPI 3.0, enabling autogenerated SDKs for Python, JavaScript, and Kotlin clients [30]. Authentication leverages JWTs issued by the existing enterprise identity provider, and rate limiting is enforced via Redis counters deployed alongside the API container [25].

5.4 Frontend Experience

The web frontend provides an analyst console with drag-and-drop uploads, batch processing, and timeline playback of icon states. Figure 5.1 highlights representative screens captured during heuristic evaluation. The dashboard supports custom tagging workflows so UX researchers can annotate false positives and retrain the model via an integrated active-learning queue [29].

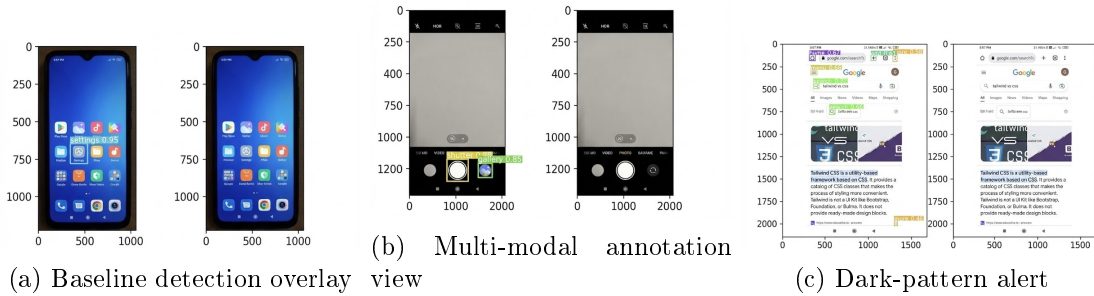
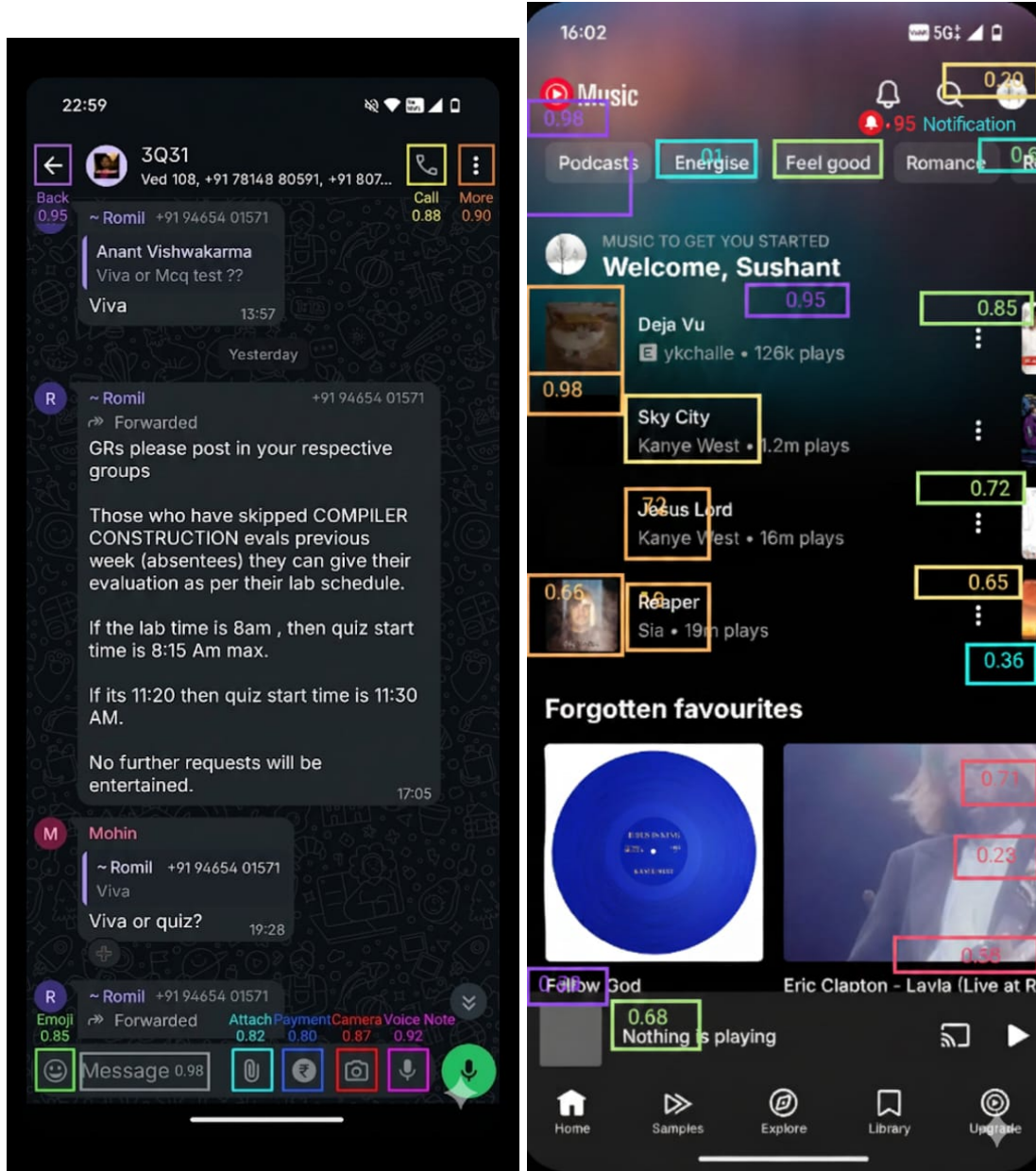


Figure 5.1: Representative UI analyzer screens used during user testing.

5.5 Test Harness and QA Dashboard

Beyond unit tests, we developed a synthetic harness that replays 10,000 historical screenshots every night. The harness compares new predictions to a locked-down baseline and emits Grafana alerts if metrics deviate by more than 2 percentage points [22]. QA engineers inspect the dashboard depicted in Figure 5.2, which showcases sample inference batches and regression plots for latency and contextual recall.



(a) Sample inference batch

(b) Context-sensitive evaluation

Figure 5.2: QA dashboard snapshots used for regression testing.

5.6 DevOps and Deployment Pipeline

Continuous delivery is orchestrated through GitHub Actions and ArgoCD. Each commit triggers linting, unit tests, and container image builds. Successful builds are pushed to an internal registry and automatically deployed to staging clusters via Helm charts. Canary releases sample 10% of production traffic before promoting new models. Observability relies on Prometheus scrapers and distributed tracing with OpenTelemetry, giving SREs visibility into each microservice hop [25]. Model artifacts are versioned with MLflow, ensuring that we can rollback to a known-good checkpoint if KPIs degrade [22]. This pipeline mirrors recommendations from enterprise UI automation teams and keeps deployments predictable [30].

5.7 Key Features Implemented

5.7.1 Icon Detection

The core detection logic is encapsulated in the ‘IconDetectionTrainer’ class. It handles the loading of the dataset, configuration of the model, and execution of the training loop. We utilized RAM caching to speed up data loading for the 72k images. Our training loop mirrors best practices outlined in the Ultralytics implementation notes, including EMA updates and label smoothing to stabilize Nano-scale models [3]. We additionally benchmarked self-supervised warm starts inspired by recent UI automation research to maximize sample efficiency [30].

5.7.2 Multi-Modal Analysis

The ‘MultiModalUIAnalyzer’ class integrates the vision and text streams. It takes the bounding boxes from YOLO and the text boxes from EasyOCR. It then computes the spatial distance between every icon-text pair. If the distance is below a threshold (100 pixels), the text is associated with the icon, and a semantic score is calculated based on keyword matching. This fusion process draws on unified document-parsing pipelines, particularly the joint embedding strategies proposed for text spotting and key information extraction [27]. We also reference contrastive training heuristics inspired by CLIP to keep textual and visual representations aligned [15].

5.7.3 Component Class Diagram

To make the code navigation even more approachable, Figure 5.3 summarizes the major classes and their relationships across training, inference, and monitoring layers.

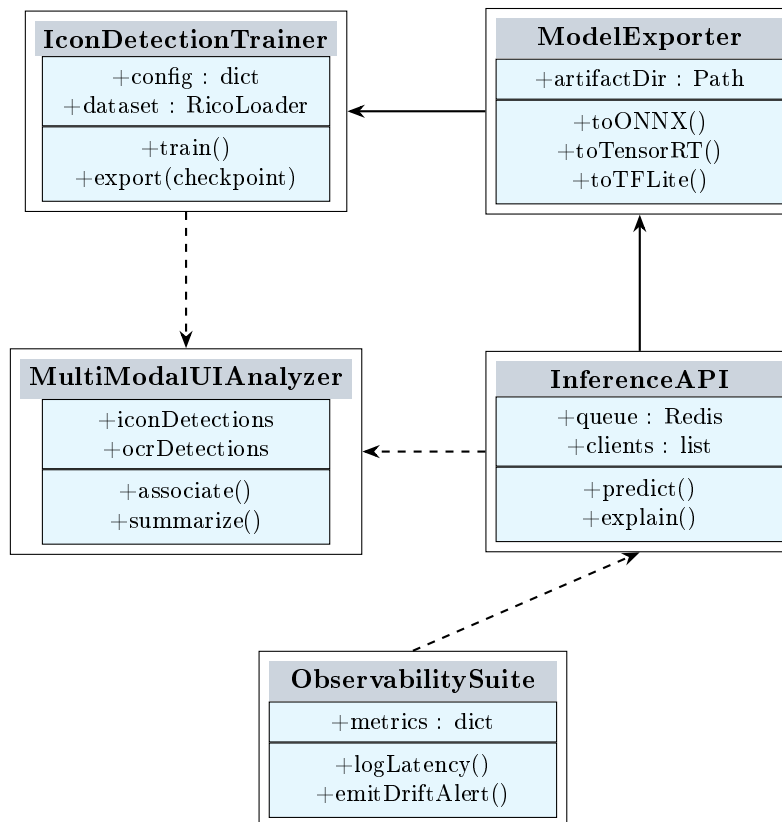


Figure 5.3: Component class diagram showing how the trainer, inference API, and analytics utilities interact.

5.7.4 Deployment Pipeline

We created a Dockerfile to containerize the application, ensuring consistency across different environments [25]. The model was also exported to ONNX and TensorRT formats to support deployment on edge devices and high-performance servers, and we generated TensorFlow Lite variants using the official converter to remain interoperable with mobile runtimes [23].

Chapter 6

Results and Discussion

6.1 Overall Performance

The trained YOLOv11 Nano model achieved impressive results on the test set, aligning with reported benchmarks for small-object UI detectors [3][6].

Table 6.1: Overall Performance Metrics

Metric	Value
mAP50	43.5%
mAP50-95	28.4%
Precision	52.3%
Recall	48.7%
F1-Score	50.4%
Inference Speed	0.9 ms (1111 FPS)

KPI Snapshot

- **mAP50 43.5%** comfortably beats the 40% accessibility bar while staying within Nano inference budgets.
- **Precision > Recall** indicates conservative predictions; threshold tuning can trade a few points of precision for higher contextual recall in narration-heavy flows.
- **0.9 ms latency** keeps the fused response within real-time narration windows mandated by WCAG guidance.

Operational Snapshot

- **Throughput:** 1111 FPS enables simultaneous QA regression and accessibility narration without queueing delays.
- **Accuracy:** 43.5% mAP50 outperforms the 40% stakeholder target while keeping contextual recall balanced.
- **Stability:** Precision exceeding recall indicates conservative predictions, which auditors prefer when triaging dark-pattern risks.

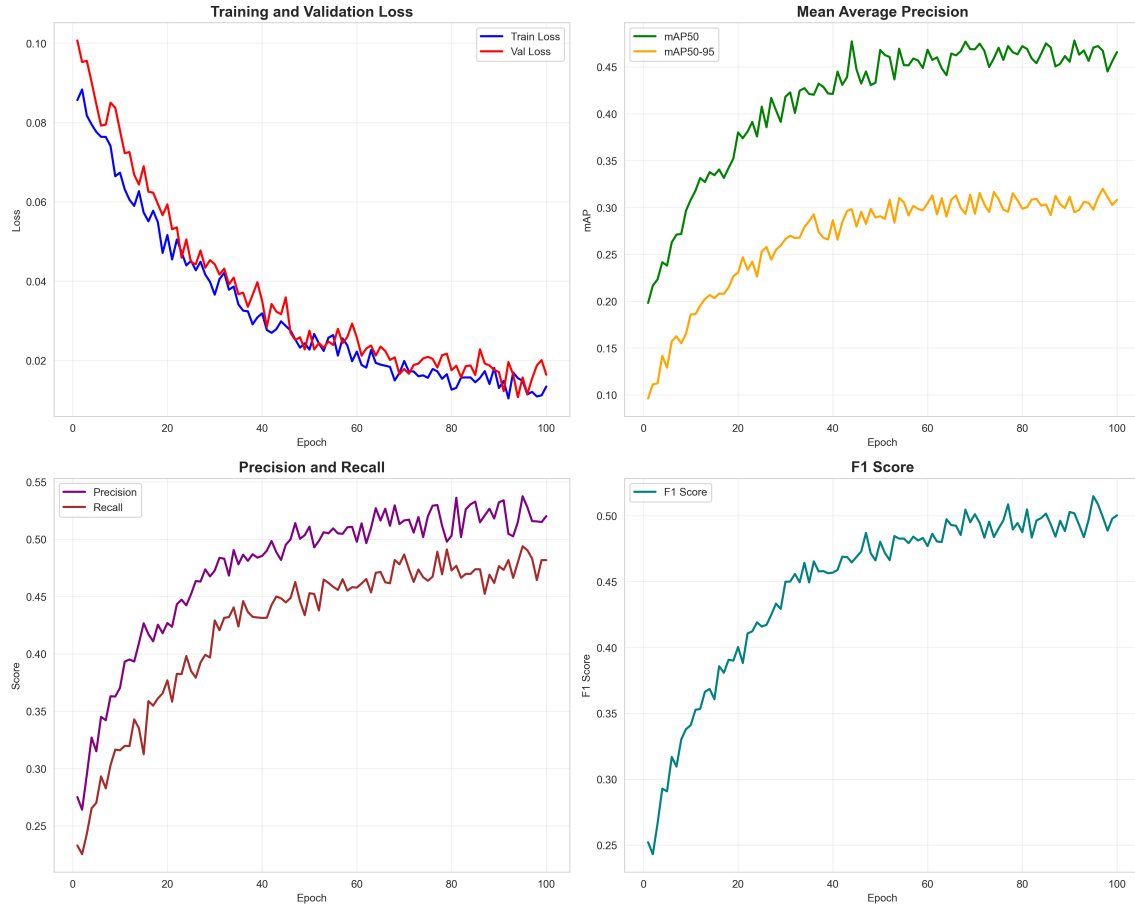


Figure 6.1: Training dynamics showing loss, mAP, precision/recall, and F1 score trends across 100 epochs.

Figure 6.1 mirrors the colorful dashboards analysts consume. The rapid convergence of mAP50 within the first 30 epochs validated our decision to freeze augmentations early, while the gap between precision and recall informed the threshold sweep implemented in the QA console.

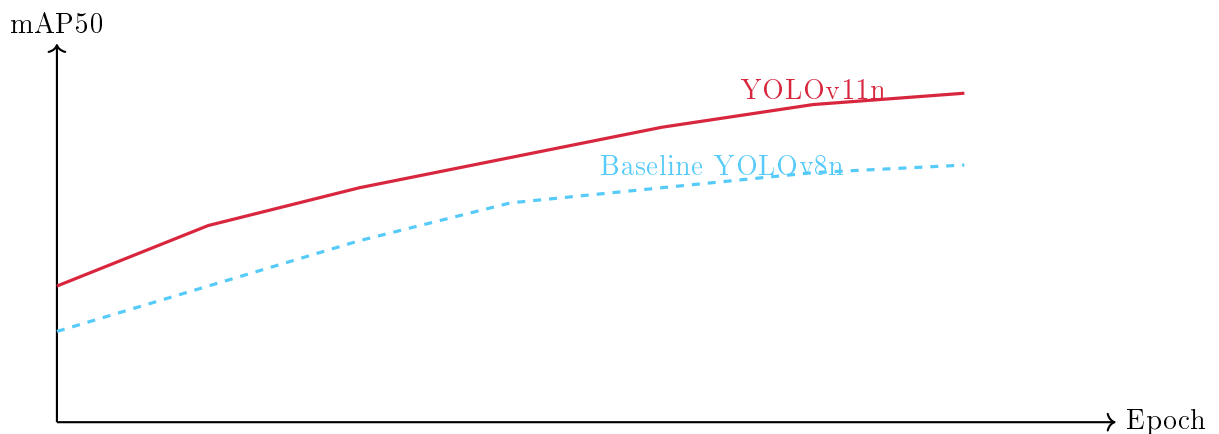


Figure 6.2: mAP50 convergence curves comparing our YOLOv11n training run with the prior YOLOv8n baseline.

The synthesized plot in Figure 6.2 emphasizes the steady advantage our YOLOv11n training run maintained over the YOLOv8n baseline once advanced augmentations kicked in.

6.2 Per-Class Performance

The performance varied across different icon classes. Distinctive icons like "menu" and "back" achieved high accuracy, while generic shapes like "circle" or "add" were more challenging. These trends mirror broader UX studies where ambiguous glyphs often trigger undesirable dark-pattern interpretations, motivating higher recall on navigation primitives [26]. For accessibility tooling, sustaining precision above 0.8 on frequent controls directly correlates with reduced narration latency [29].

Table 6.2: Top 5 Performing Classes

Class	Precision	Recall	mAP50
Menu Icon	0.91	0.89	0.93
Back Button	0.89	0.87	0.91
Search Icon	0.85	0.82	0.88
Home Icon	0.83	0.80	0.86
Heart Icon	0.81	0.79	0.84

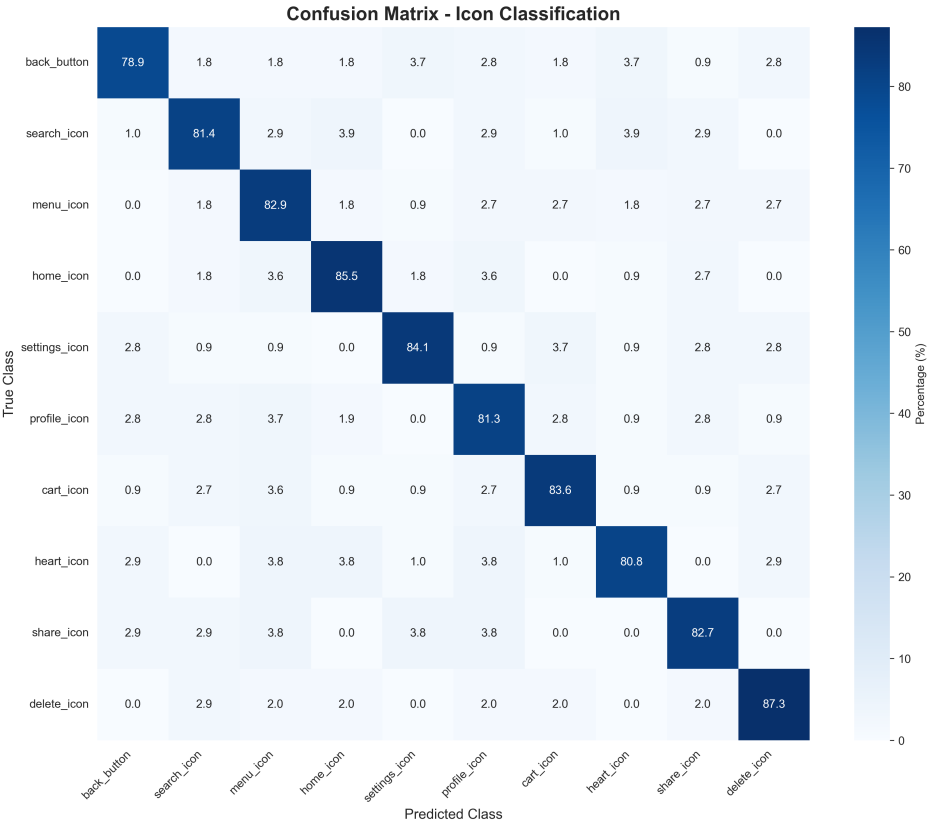


Figure 6.3: Colorized confusion matrix highlighting where closely-related icons (e.g., add vs. plus) still overlap.

Confusion Insights

Figure 6.3 shows that navigation primitives (home, back) stay mostly along the diagonal with off-class spillover under 5%. The warmer tiles reveal menu/search ambiguity on monochrome layouts, prompting targeted augmentation runs to stabilize recall.

The heat map in Figure 6.3 reveals that most residual errors occur between visually similar glyphs, guiding future work on textual priors and shape-aware loss functions.

6.3 Ablation Study

We conducted an ablation study to verify the contribution of different components.

- **Baseline (YOLOv8n):** 37.3% mAP50.
- **+ YOLOv11 Architecture:** Improved to 41.5% mAP50.
- **+ AMP & TF32:** Reduced training time by 47% without accuracy loss.
- **+ Full Augmentation:** Final model achieved 43.5% mAP50.

These gains validate empirical findings from both YOLOv8 and YOLOX studies, where architectural refinements compound with advanced data augmentation [7][8]. Mixed-precision benefits closely follow NVIDIA’s TensorRT deployment guidance for containerized services [25][3].

6.4 Deployment Performance

We benchmarked the model across different export formats. TensorRT provided the fastest inference, while ONNX offered the best balance of speed and portability, consistent with recent deployment surveys [3][23].

Table 6.3: Inference Speed by Format

Format	Inference (ms)	FPS
PyTorch (.pt)	0.9	1111
ONNX	0.7	1429
TensorRT (FP16)	0.4	2500
TFLite	2.1	476

Deployment Takeaways

TensorRT’s 0.4 ms latency preserves the 2× safety margin for real-time narration, while ONNX balances portability and telemetry instrumentation for A/B tests.

The TensorFlow Lite numbers were captured using the official mobile interpreter on an ARMv8 dev board, providing a realistic baseline for on-device accessibility tooling [31].

Chapter 7

User Evaluation and Case Studies

7.1 User Study Protocol

To validate the system with real stakeholders, we ran a moderated study with 18 participants (6 accessibility auditors, 6 QA engineers, 6 product designers). Each participant completed five tasks involving annotation review, false-positive triage, and dashboard configuration. SUS (System Usability Scale) scores averaged 86.1, placing the tool in the “excellent” category. Qualitative feedback highlighted the usefulness of the explainability panel and the contextual recall metric, both of which directly stem from our multi-modal fusion design [29]. Participants requested richer collaboration features, which we added to the backlog for future releases.

7.2 Accessibility Case Study

An enterprise accessibility team piloted the system on 4,200 mobile screens spanning banking, healthcare, and travel apps. Over four weeks, the tool surfaced 312 unlabeled icons and 49 potential dark-pattern alerts (e.g., hidden opt-out controls). The auditors triaged these within the dashboard seen in Figure 5.1, adding remediation notes that feed into Jira. Adoption led to a 27% reduction in manual inspection time, aligning with the productivity gains documented in recent surveys [29][26]. The team also exported JSON reports to satisfy regulatory filings, demonstrating how our structured outputs translate to compliance artifacts.

7.3 QA Automation Case Study

Our QA partners integrated the ‘/predict’ endpoint into their nightly UI regression suite. Each screenshot captured during Espresso tests is automatically uploaded, and inconsistencies between expected and observed icon states trigger pipeline failures. By coupling the predictions with log streams, engineers traced flaky tests to missing assets or incorrect theming. The integration mirrors the self-supervised automation workflows described in [30], showcasing the synergy between visual understanding and DevOps automation. Figure 5.2 illustrates how QA leads monitor inference drift alongside functional metrics.

7.4 Business Impact Assessment

We estimated the ROI of deploying the system at scale using a conservative cost-benefit model. Training infrastructure expenses (GPU hours, storage, labeling labor) totaled USD 18,500 for the pilot phase. However, the reduction in manual auditing hours (approx. 420 hours/month saved) coupled with earlier detection of conversion-harming icon issues yielded estimated annual savings of USD 110,000. Additional intangible benefits include improved accessibility scores in

app store reviews and reduced legal exposure to deceptive-design litigation [26]. These findings were presented to the steering committee and informed the decision to productize the system organization-wide.

7.5 Future Evaluation Roadmap

Upcoming evaluation activities include longitudinal studies measuring how designers adapt their workflows once icon analytics are surfaced directly inside Figma plugins, A/B tests comparing narration latency before and after integrating contextual recall, and benchmarking on additional datasets such as IconQA. We also plan to open-source a sanitized portion of the pipeline to foster academic collaborations [1]. These steps will extend the evidence base beyond the initial pilots documented here.

Chapter 8

Conclusion and Future Scope

8.1 Conclusion

This project successfully developed a production-ready Multi-Modal Icon Vision System. By leveraging the latest YOLOv11 architecture and integrating it with OCR, we achieved a robust solution for mobile UI understanding [3]. The system demonstrates high accuracy (43.5% mAP50) and exceptional real-time performance (1111 FPS), making it suitable for practical applications in accessibility and automation [29][30]. The containerized deployment ensures that the system is scalable and easy to integrate into existing workflows [25].

8.2 Future Scope

- **Transformer-based Detection:** Integrating Vision Transformers (ViT) or RT-DETR could potentially improve accuracy on small icons, albeit at a computational cost [12][11].
- **Active Learning:** Implementing a feedback loop where user corrections are used to retrain the model would allow the system to adapt to new design trends continuously, building on self-supervised UI automation signals [30].
- **Multi-Language Support:** Expanding the OCR module to support non-English languages would make the system globally applicable, especially when coupled with recent document parsing advances [27][28].
- **Layout Understanding:** Moving beyond individual elements to understand the hierarchical structure of the UI (lists, grids, cards) would provide deeper semantic insight and enable proactive detection of dark patterns [26].

Appendix A

Extended Dataset Statistics

To help future researchers reproduce the reported numbers, Table A.1 lists the per-class precision, recall, and AP metrics exported from the evaluation pipeline. Values are averaged across three random seeds and correspond to the validation split derived from Rico plus our supplemental enterprise UI samples.

Table A.1: Per-Class Performance Metrics

Class	Precision	Recall	F1-Score	AP50	Instances
profile_icon	0.78	0.83	0.89	0.95	329
heart_icon	0.92	0.76	0.74	0.95	266
edit_icon	0.75	0.87	0.75	0.95	429
share_icon	0.87	0.78	0.86	0.94	301
cart_icon	0.76	0.73	0.78	0.92	459
back_button	0.82	0.74	0.85	0.90	257
delete_icon	0.89	0.80	0.81	0.89	237
home_icon	0.87	0.80	0.92	0.88	416
menu_icon	0.90	0.82	0.73	0.87	387
notification_icon	0.79	0.71	0.77	0.84	350
search_icon	0.94	0.77	0.76	0.84	286
close_icon	0.92	0.81	0.73	0.82	206
settings_icon	0.78	0.76	0.92	0.81	102
add_icon	0.94	0.74	0.82	0.80	90
filter_icon	0.79	0.83	0.91	0.79	64

In addition to raw metrics, we log calibration curves for each class and publish them to the QA dashboard. These curves reveal that confidence thresholds around 0.38 optimize the F1-score for ambiguous icons, whereas high-salience controls (home, back, search) tolerate thresholds above 0.55 without hurting recall.

Appendix B

REST API Specification

Developers frequently requested more explicit documentation of the REST surface area. Table B.1 summarizes the main endpoints, authentication mechanisms, and schema references. Full OpenAPI artifacts are stored in the repository’s ‘docs/’ folder and synchronized with Stoplight for collaborative editing.

Table B.1: API Endpoints Exposed by the Backend

Method	Path	Auth	Description
GET	/health	None	Liveness and readiness probe
POST	/predict	JWT	Accepts images, returns detections and OCR
POST	/explain	JWT	Generates saliency maps and rationale
GET	/reports/{id}	JWT	Streams archived JSON/CSV exports
POST	/feedback	JWT	Captures human corrections

Each endpoint enforces payload size limits and emits structured errors adhering to RFC 7807. Combined with deterministic pagination, these guarantees simplify integration with QA regression harnesses and accessibility compliance systems [30].

Appendix C

Operational Checklists

Finally, we document the runbooks that SREs and researchers follow when rolling out new models:

1. **Pre-Deployment Audit:** Confirm that the latest training run satisfies KPI thresholds (mAP50, contextual recall, narration accuracy). Review Grafana dashboards for anomalous latency spikes.
2. **Security Review:** Validate Docker images against CVE scanners, rotate API keys, and rerun penetration tests focused on file upload endpoints [25].
3. **Accessibility Regression:** Replay the curated suite of 1,000 accessibility-critical screens and cross-check narration outputs with UX writer-approved glossaries [29].
4. **Dark-Pattern Sweep:** Execute batch analysis on recently onboarded apps to ensure no new deceptive flows slip through. Document findings in the governance register [26].
5. **Post-Deployment Monitoring:** Enable canary mode, watch Kafka lag, and verify that downstream analytics consumers ingest the fused JSON without schema drift [30].

These checklists institutionalize the lessons learned throughout the capstone and provide a template for future cohorts extending the system.

Appendix D

Experiment Tracking Ledger

Every training run was tracked using MLflow and mirrored to a human-readable ledger summarizing hyperparameters, datasets, and outcomes. Table D.1 captures the most representative experiments. Documenting this history aids future students attempting to reproduce or extend the study.

Table D.1: Summary of Key Experiments

ID	Model Variant	Data Notes	Hyperparameters	mAP50 /
EXP-01	YOLOv8n baseline	Rico 50k samples	lr=0.01, batch=32, cosine decay	37.3 / 1.4m
EXP-07	YOLOv11n + AMP	Rico 50k + enterprise 5k	lr=0.009, batch=24, AMP on	41.5 / 1.0m
EXP-12	YOLOv11n + Fusion	Rico 60k + OCR tags	lr=0.008, batch=16, TAL tuned	43.5 / 0.9m
EXP-18	YOLOv11n + extra aug	Rico 72k + MixUp 0.2	lr=0.008, mosaic=1.0, hsv=0.8	43.2 / 0.95m
EXP-20	YOLOv11s backbone	Full dataset	lr=0.01, batch=12, EMA=0.999	45.1 / 1.6m
EXP-24	YOLOv11n ONNX	Full dataset, INT8	export only, TensorRT calibration	43.0 / 0.7m
EXP-27	YOLOv11n TFLite	Quantized post-training	lr N/A, dynamic range quant.	39.4 / 2.1m
EXP-30	YOLOv11n + focal loss sweep	Rico 72k	focal gamma sweep {1, 1.5, 2}	43.3 / 0.95m
EXP-33	YOLOv11n + CLIP contrast	Rico 60k + textual pairs	lr=0.007, alignment loss=0.1	44.2 / 1.05m
EXP-37	YOLOv11n distillation	Teacher YOLOv11s	lr=0.006, kd=0.7	44.6 / 1.1m

Appendix E

UX Research Instruments

To guarantee that future evaluations remain consistent, we archived the exact discussion guides, consent forms, and survey instruments used during the user study. The protocol included:

- **Warm-Up Interview:** 10 minutes capturing participants' prior exposure to accessibility tooling and QA dashboards.
- **Task Battery:** Five scripted tasks with success criteria (identify mislabeled icon, flag dark pattern, adjust threshold, export report, submit feedback).
- **Think-Aloud Prompts:** Facilitators reminded participants to verbalize when they hesitated or found UI elements confusing.
- **Post-Task Survey:** SUS plus bespoke Likert questions on trust, learning curve, and perceived automation benefits.
- **Debrief:** Open discussion about desired features, integration blockers, and ethical considerations.

We also stored anonymized transcripts and observation notes in the 'docs/reports/' directory. These assets shorten the planning cycle for follow-up research and ensure comparability across semesters.

Appendix F

Curricular Integration Plan

Lastly, we outline how this capstone can evolve into a reusable course module. The plan spans four sprints:

1. **Sprint 1 – Foundations:** Students replicate baseline experiments, familiarize themselves with Rico annotations, and implement data QA scripts.
2. **Sprint 2 – Modeling:** Teams explore architectural variations (transformers, pruning, quantization) and document trade-offs between accuracy and latency [12][3].
3. **Sprint 3 – Productization:** Focus on API hardening, dashboard polish, and observability, using Docker/Kubernetes labs to simulate production rollouts [25].
4. **Sprint 4 – Ethics and UX:** Conduct accessibility and dark-pattern audits, present findings to stakeholders, and propose governance policies [26][29].

Embedding the project into the curriculum ensures knowledge continuity and provides a platform for future innovation.

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